SCHOOL YEAR 2025-2026

COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING

I. **COURSE CODE** : **IE-402**

II. **COURSE TITLE** : **Systems Engineering**

III. COURSE DESCRIPTION :

This course introduces students to the principles and practices of systems engineering, focusing on the analysis, design, and management of complex systems. Through a blend of theoretical concepts and practical applications, students learn how to approach and solve large-scale engineering problems involving multiple components, stakeholders, and operational constraints. Key topics include requirements analysis, systems design, integration, project management, and performance evaluation.

Students will develop skills in modeling and optimizing systems that integrate people, processes, and technology. They will explore methods for ensuring that systems are efficient, reliable, and adaptable across various industries, such as manufacturing, healthcare, and logistics. This course prepares BSIE students to play a vital role in creating systems that maximize performance, minimize waste, and support organizational goals.

Beyond technical knowledge, the course integrates Marian Augustinian values into each lesson, fostering the development of students as Christo-centric Catholic professionals. The curriculum not only aims for intellectual growth but also instills a commitment to ethical decision-making, social responsibility, and compassionate service. Students will be equipped with the knowledge and values necessary to thrive as professionals in the field of Industrial Engineering.

IV. PRE – REQUISITE : 4th Year Standing

V. CREDIT UNITS : 3 units

VI. COURSE LEARNING OUTCOMES :

Vision	Mission
Moved by the Spirit of the Risen Christ, La Consolacion University Philippines, envisions itself to become a catalyst for the wholistic transformational development of Catholic Augustinian Marian graduates rooted in Gospel values.	To achieve the above vision, LCUP commits itself to generate and make available to its Augustinian Marian students, programs and services that will make them: ● *exhibit academic and technical competence in their respective areas of discipline in the local and international environment;* ● *demonstrate their knowledge, critical thinking and creativity by exploring the various avenues of learning and searching for truth;* ● *manifest their willingness to serve specially the marginalized sector of our society;* ● *demonstrate productivity in various areas of learning as responsible citizens of the society; and* ● *live the Gospel values in the varied aspects of their personal and professional lives*

EXPECTED GRADUATE ATTRIBUTES	PROGRAM OUTCOMES	COURSE LEARNING OUTCOMES
EGA 1: Critical and Creative thinker EGA 2: Competent Catholic Augustinian-Marian Professional EGA 3: Socially Responsive Steward EGA 4: Transformative Lifelong learner	PO1. Ability to apply knowledge of mathematics and science to solve complex engineering problems; PO2. Ability to design and conduct experiments, as well as to analyze and interpret data; PO3. Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability, in accordance with standards.	CLO 1: Demonstrate knowledge and understanding of engineering and management principles by effectively leading and collaborating within teams, managing projects, and engaging in multidisciplinary environments. [EGA 1] [PO 1,3] CLO 2: Apply critical and creative thinking skills to analyze, design, and optimize complex systems, integrating various components such as people, materials, technology, and information. [EGA 1] [PO 2,3, 4, 5]

	PO4. Ability to function on multidisciplinary teams PO5. Ability to identify; formulate, and solve complex engineering problems; PO6. Understanding of professional and ethical responsibility. PO7. Ability to communicate effectively PO8. Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context. PO9. Recognition of the need for, and an ability to engage in life-long learning; PO10. Knowledge of contemporary issues; PO11. Ability to use techniques, skills and modern engineering tools necessary for engineering practice; and; PO12. Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in PO13: Ability to design, develop, implement and improve integrated system that include people	CLO 3: Develop the ability to identify and solve systems engineering challenges, making decisions that improve performance, enhance efficiency, and support sustainable practices across diverse industries. [EGA 1, 2] [PO 2, 3. 5, 6,7] CLO 4. Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [EGA 2] [PO 4,5]

	materials, implementation, equipment and energy.	

VII. LEARNING PLAN:

SCHEDULE	INTENDED LEARNING OUTCOMES	COURSE CONTENTS	SUGGESTED LEARNING ACTIVITIES	SUGGESTED ASSESSMENT TASKS	EXPECTED VALUE/S TO BE INTEGRATED
Week 1	ILO10: Demonstrate the Augustinian, Marian and missionary Christ-centered Spirituality through value webbing. [CLO 4]	**I. COURSE ORIENTATION** A. Introduction to the Course Contents B. Course Requirements and Activities C. Course Objectives D. Grading System and Policies on Absences, Tardiness and Examinations	• Setting and sharing of Expectations from the Course • Discussion on the Vision, Mission and Goals of LCUP and of CITE • Presentation of Course Objectives and Contents, Course Requirements and Activities • Review of Policies on Absences, Tardiness, Examinations, Grading System	• Statement of Expectations • Review of Course Contents, Objectives, Activities and Assessment Tools • Review of Policies on Absences, Tardiness, Examinations, Grading System	• Interiority • Respect • Obedience
Week 2-3	ILO1: Describe fundamental principles of systems engineering, including system life cycles, requirements	**Introduction to Systems Engineering** • Introduction to Systems Engineering	**Lecture and Case Study Analysis** **Activity**: Present a lecture on the basics of systems engineering management, followed by	**Quiz on Fundamental Concepts** Conduct a quiz covering key concepts and terminology in	• Understanding and patience in learning • Creative and critical thinking

	analysis, and systems thinking. [CLO 1], [CLO 2] ILO2: Develop basic system designs that incorporate requirements from multiple disciplines, including human factors, technology, and organizational needs. [CLO 1], [CLO 2]	Management • Systems Engineering Management in DoD Acquisition	an analysis of a real-world case study in systems engineering, such as a notable DoD acquisition project. **Interactive Discussion and Q&A** **Activity**: Facilitate an interactive class discussion on the key principles of systems engineering management and its importance in large-scale projects, especially in high-stakes environments like defense.	systems engineering management, including its principles, life cycle stages, and specific applications in DoD acquisition. **Case Study Analysis Report** Provide a case study on a DoD acquisition project or a similar large-scale systems engineering project. Students will analyze and write a report covering the systems engineering management challenges and solutions highlighted in the case.	• Analytical thinking • Good active participation and involvement • Appreciation • Effective and efficient management of resources • Appreciation • Awareness • Courage and Humility
WEEK 4	[ILO 1], [ILO 2]	**PRELIMINARY EXAM**	Examination covering the topics discussed	Examination results	Courage and Humility
WEEK 5-8	ILO3: Summarize the overall systems engineering process, explaining each phase from requirements analysis to verification and process outputs. [CLO1][CLO2]	**The Systems Engineering Process** • Systems Engineering Process Overview • Requirements Analysis • Functional Analysis and Allocation • Design Synthesis	**Group Workshop on Functional Analysis and Allocation** **Activity**: In small groups, students will break down system requirements into functional components and allocate these to hypothetical subsystems. **Group Workshop on Functional Analysis**	**Requirements Analysis Report** Assign students a hypothetical project scenario where they must identify, analyze, and document key system requirements based on provided stakeholder needs.	• Understanding and patience in learning • Creative and critical thinking • Analytical thinking • Good and active

	ILO4: Analyze and define system requirements based on stakeholder needs, constraints, and project objectives. [CLO2][CLO3] ILO5: Recognize key outputs of the systems engineering process, such as documentation, specifications, and test reports, and evaluate their role in supporting successful project outcomes. [CLO2][CLO3]	• Verification • Systems Engineering Process Outputs	**and Allocation** In small groups, students will break down system requirements into functional components and allocate these to hypothetical subsystems. **Reflection Assignment on Process Outputs** Have students reflect on the outputs generated at each stage of the systems engineering process and discuss how these support project success.	**Functional Analysis and Allocation Diagram** **Description**: Have students create a functional flow diagram for a system based on a given set of requirements, breaking down functions and allocating them to subsystems. **Design Synthesis Project** **Description**: Assign a design synthesis task where students create an initial design for a system, explaining how the design integrates functional requirements and meets project goals.	participation and involvement • Appreciation • Effective and efficient management of resources • Problem solving skills •
WEEK 9	[ILO 3], [ILO 4] [ILO5]	**MIDTERM EXAM**	• Examination covering the topics discussed	• Examination results	• Skills and Competencies • Critical Thinking • Honesty
Week 10-11	ILO6: Create and interpret a WBS to organize and manage project components, identifying key deliverables and tasks for effective project planning and	**Systems Analysis and Control** • Work Breakdown Structure • Configuration Management	**Interactive Workshop on Work Breakdown Structure (WBS) Creation** **Activity**: Conduct an interactive workshop where students create a WBS for a hypothetical	**Peer Review of Experimental Design and Robustness Plan** Ask students to peer review each other's experimental design proposals and	• Understanding and patience in learning • Creative and critical thinking • Analytical thinking • Good and

	control. [CLO 3] [CLO4] ILO7: Apply modeling and simulation techniques to analyze system behavior, test potential solutions, and predict outcomes in a controlled virtual environment. [CLO 3] [CLO4] ILO8:Identify, assess, and manage risks within a systems project, developing risk mitigation strategies to minimize potential negative impacts on project outcomes. [CLO 3] [CLO4]	• Technical Reviews and Audits • Trade Studies • Modeling and Simulation • Metrics • Risk Management	project, breaking down tasks and identifying deliverables. **Case Study on Configuration Management** **Activity**: Present a case study demonstrating configuration management principles and practices in a real-world project. Have students discuss the role of configuration management in maintaining system consistency. **Risk Management Analysis Activity** **Activity**: Present a project scenario and ask students to identify potential risks, assess their impact and probability, and develop mitigation strategies.	robustness plans, providing constructive feedback on methodology and approach. **Risk Assessment and Mitigation in Robust Design** Assign a task where students assess potential risks to robustness in a system design and develop mitigation strategies to address these risks.	active participation and involvement • Appreciation • Effective and efficient management of resources • Problem solving skills
Week 12-14	ILO9: Formulate a comprehensive systems engineering plan that outlines objectives, resources, timelines, and milestones to guide system development from inception	**Planning, Organizing and Managing** • Systems Engineering Planning • Product Improvement Strategies	**Workshop on Systems Engineering Planning** Conduct a hands-on workshop where students outline a systems engineering plan for a hypothetical project. They should include project objectives, resource	**Systems Engineering Plan Development** **Task**: Have students create a comprehensive systems engineering plan for a hypothetical project. This plan	• Active participation and involvement • Active participation and involvement • Critical

	through completion. [CLO 3] [CLO4] ILO10: Evaluate essential management considerations—such as resource allocation, risk assessment, and stakeholder engagement—and their impact on project success, applying these principles to real or simulated project scenarios. [CLO 3] [CLO4]	• Organizing and Integrating System Development • Contractual Considerations • Management Considerations and Summary	allocation, timeline planning, and key milestones. **Product Improvement Strategy Brainstorming Session** In groups, students brainstorm and propose strategies for iterative product improvement in a given scenario, focusing on design, functionality, and customer feedback loops.	should include objectives, resource allocation, timeline, milestones, risk assessment, and project phases. **Product Improvement Strategy Proposal** **Task**: Ask students to develop a proposal for a product improvement strategy for an existing system, identifying areas for enhancement and proposing solutions to improve quality, efficiency, and user satisfaction.	Thinking • Patience • Understanding • Critical Analysis • Logical Thinking
Week 15-17	ILO11: Analyze and evaluate multiple design alternatives through trade space exploration, applying criteria to select the optimal concept based on performance, feasibility, and stakeholder needs. [CLO 3] [CLO4] ILO 12: Develop and execute experiments to validate design	**Application** • Systems Engineering As Human Activity • Trade Space Exploration Concept Selection • Experiments • Robust Design I • Robust Design II	**Case Study Discussion: Systems Engineering as a Human Activity** Provide a case study that highlights the role of human factors in systems engineering, including team dynamics, stakeholder involvement, and ethical considerations. Facilitate a class discussion on the human-centered aspects and their impact on project outcomes.	**Robust Design I – System Evaluation and Improvement Proposal** **Task**: Have students apply basic robust design techniques to evaluate a system and propose design changes that would make it more stable under varying conditions. They should identify critical factors that affect performance and	• Skills and Competencies • Critical Thinking • Honesty

	assumptions, analyze system behaviors, and refine solutions, demonstrating an understanding of experimental design and data collection principles. [CLO 3] [CLO4] ILO 13: Identify critical factors in system design that influence performance consistency, and apply robust design techniques to improve reliability under varying conditions (Robust Design I). [CLO 3] [CLO4] ILO14: Utilize advanced robust design methods to enhance system resilience, focusing on minimizing performance variability in complex, multi-variable systems (Robust Design II). [CLO 3] [CLO4]		**Design and Conduct Experiments Activity** Assign students a mini-project to design and conduct an experiment to validate a system component or process. They should identify the objective, develop a methodology, collect data, and analyze results to inform design improvements. **Robust Design Brainstorming Session (Robust Design I)** In small groups, students brainstorm how to apply basic robust design principles to make a system more stable under varying conditions. They discuss factors that influence system stability and propose design adjustments to minimize variability. **Advanced Robust Design Project (Robust Design II)** **Activity**: For a more complex project, students apply advanced robust design techniques to a hypothetical or real-world system. They identify and	suggest improvements to minimize variability. **Robust Design II – Advanced Design Analysis** **Task**: For a more complex system, students should apply advanced robust design methods to assess and improve the system's performance. They will use tools such as design of experiments (DOE) to explore multiple design variables and their interactions. **Robust Design Report (I & II)** **Task**: Students will submit a report for each part of the robust design application, detailing the steps taken to assess and improve system performance. For Robust Design I, they will focus on basic principles, and for Robust Design II, they will apply	

			control multiple variables to minimize performance variation across diverse scenarios.	advanced techniques to optimize the system.	
Week 18	[CLO 3] [CLO 4]	**FINAL EXAMINATION**	• Examination on the topics discussed during the Final Term		• Skills and Competencies • Critical Thinking • Honesty

VIII. COURSE REQUIREMENTS :

1. Class attendance
2. Recitations
3. Quizzes
4. Examinations
5. Compilation of activity reports

IX. REFERENCES

A. BOOKS

Awari, G. K., and Warjurkar, S. V. (2022). *Ethics in information technology: a practical guide.* CRC Press. TK 5105.878 A93 2022

Bernard, M. (2022). *Future skills: the 20 skills and competencies everyone needs to succeed in a digital world.* John Wiley & Sons, Inc. T 65.3 M37 2022

Delfanti, A. (2019). *Introduction to digital media.* Wiley. QA 76.575 D48 2019

B. INTERNET REFERENCES

MIT Open Courseware (2024). "Introduction to Design Computing" https://ocw.mit.edu/courses/4-500-introduction-to-design-computing-fall-2008/download/ Massachusetts Institute of Technology

C. JOURNALS

Primiero, G. (2019). Introduction." *On the Foundations of Computing*, pp. 1–4., doi:10.1093/oso/9780198835646.003.0001.

Jianwen, C. and Wakil, K. (2019), "A model for evaluating the vital factors affecting cloud computing adoption: Analysis of the services sector", *Kybernetes*, doi:10.1108/K-06-2019-0434

Yang, F., Akanbi, T., Chong, O. W., Zhang, J., Debs, L., Chen, Y., & Hubbard, B. J. (2024). Project-Based Introduction to Computing in Construction Management Curriculum: A Case Study. *Journal of Civil Engineering Education*, *150*(1). https://doi.org/10.1061/JCEECD.EIENG-1905

X. GRADING SYSTEM

1. Class Standing 60%
2. Examination 40 %
3. Term Grade CS (60%) + TE (40%)
4. Final Rating $\frac{\text{Prelim Grade + Midterm Grade + Final Grade}}{3}$

Prepared by:	Checked and Reviewed by:	Approved by:
ENGR. GLAZEN CAREY PEREZ Faculty/Date	**EDNA F. DAYAO, DIT** Dean/Date **ROSALYN S. GALVEZ, PhD., LPT** Asst. Vice President for Academic Affairs-Date References Certified by: **DR. MARIA MERCEDEZ M. HAZ** LIC Director/Date	**JAY MARK B. SANTOS, PhD, LPT, CSSWB** Vice President for Academic Affairs-Date

Revision Date:	
Modifications: __Learning Outcomes __Course Contents __Suggested Learning Activities __Suggested Assessment __Expected Values to be Integrated __Course Requirements __References __Grading System	